



Finish the Stack

Add the missing block to reach the star

Name:

Date:

★ I can choose the missing block that gets Bit to the star.

One block is missing

Each program has one empty block. Pick the move block that gets Bit to the star ★. Run the whole stack from the top to check where Bit lands.

Stage 1 — star on box 4



Program A



1 Finish Program A

Bit starts on box 1. After → → Bit is on box 3. Which block finishes the program so Bit lands on the star (box 4)? Circle → or ←, then draw it in the empty block.

Program B



2 Finish Program B

The empty block is in the MIDDLE this time. Which block makes Bit land on the star (box 4)? Circle ➡ or ⬅, then draw it in the empty block.

Stage 2 — Bit on box 4, star on box 2



Program C



3 You try — Program C

Now Bit starts on box 4 and the star is on box 2, to the LEFT. After ← Bit is on box 3. Which block finishes the program so Bit lands on the star? Circle → or ←, then draw it in the empty block.

Big idea

To finish a program, pick the block that makes it do the job. Always run the whole stack to check Bit lands where you want.



Finish the Stack

Quick facilitation notes & answer key

What this worksheet teaches

Children complete a stack by choosing the one move block that makes Bit reach the star, then run the whole stack to check. Picking the block that gives the result you want is early program design. No coding experience needed.

Before you start (no computer needed)

No computer — paper, a pencil, and a crayon. You read each prompt aloud. Optional: paper blocks (a green flag block + arrow blocks) to stack by hand, or floor boxes the child can walk.

How to lead it (about 15 minutes)

1. Show them (you do). Run a stack with one block missing and ask: "which block lands Bit on the star?"
2. Do one together. Exercise 1 — the missing last block is a right arrow (Bit reaches box 4).
3. Let them try. Children fill the middle block (Ex 2) and a left-arrow ending (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — the missing block is a right arrow (box 1 → 2 → 3 → 4, the star). Exercise 2 — the missing middle block is a right arrow (box 1 → 2 → 3 → 4).

Exercise 3 — the missing block is a left arrow (box 4 → 3 → 2, the star).

If children get stuck — and ways to adjust

Watch for: choosing the right number of steps but the wrong direction. Check which way the star is first.

Make it easier: run the blocks that are there, then ask "one more step to where?"

Make it harder: ask for a different stack that also reaches the star.

Make it active: walk the stack on floor boxes and test the chosen block.

Ontario Kindergarten link (official wording)

Problem Solving and Innovating — children select and test instructions to reach a goal.

In plain terms: choosing the one block that makes Bit reach the star.

No reading is needed from the child — you read each prompt aloud; the child circles, numbers, or draws an arrow.

You'll know it worked when

The child chooses the move block that lands Bit on the star and checks it by running the stack.